GLOBAL MUSIC STREAMING LISTENER PREFERENCES

Exploratory Data Analysis (EDA) | Python Project

A concise project brief for recruiters and hiring teams

PROJECT AT A GLANCE

5,000	10	6	0
listener records	countries	streaming platforms	duplicate rows reported

Project summary

This project explores a global music-streaming listener dataset to understand listening habits, platform usage, genre preferences, artist preferences, and engagement patterns. The notebook uses Python for data inspection and exploratory analysis, with visualizations designed to make patterns easier to interpret.

Tools used: Python, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook.

Prepared from the submitted project notebook. The report describes the analyses present in the notebook and does not claim business experiments, predictive modelling, or causal conclusions.

1. Business Context & Objectives

Music platforms compete on discovery, personalization, and repeat engagement. Understanding who listens, what they listen to, when they listen, and how engagement varies can help frame better product and marketing questions.

Objectives

• Profile the listener base by age and country.

• Explore usage across streaming platforms and preferred music genres.

• Review listening-time patterns (morning, afternoon, or night).

• Compare the most-played artists across Free and Premium subscription types.

• Explore how music genre distributions vary across age groups.

• Inspect data quality through missing-value, duplicate, and category checks.

2. Dataset Overview

The notebook loads a CSV named **Global_Music_Streaming_Listener_Preferences.csv**. It contains 5,000 records and fields covering demographics, listening behaviour, platform and genre preferences, subscription type, and engagement indicators.

Field group	Example fields in the dataset	Analytical use
Demographics	Age, Country	Listener profile and segment comparisons
Platform & taste	Streaming Platform, Top Genre, Most Played Artist	Platform, genre and artist preference exploration
Listening behaviour	Minutes Streamed Per Day, Listening Time	Usage intensity and time-of-day patterns
Engagement	Number of Songs Liked, Discover Weekly Engagement (%), Repeat Song Rate (%)	Descriptive engagement indicators
Subscription	Subscription Type	Free vs Premium comparisons

3. Data Quality & Preparation

The notebook checks missing values, duplicate rows, available countries, column names, categorical features, and unique category values. It also removes the **User_ID** field before visual analysis because the identifier is not directly useful for the planned preference comparisons.

Checks visible in the notebook: the duplicate-row check returns 0; the dataset has 5,000 observations for the numeric fields summarized. The notebook includes a missing-value check, but this report does not assert a complete null-count result for every field.

4. Exploratory Analysis Performed

Analysis	Method in notebook	Question it helps answer
Age distribution	Histogram with KDE	How is the listener age profile distributed?
Platform popularity	Count plot ordered by platform frequency	Which platforms have more records in this dataset?
Genre preferences	Genre frequency and pie chart	How are reported top genres represented?
Listening time	Count plot by morning/afternoon/night	When do listeners report listening?
Artist vs subscription	Top-artist count plot split by Free/Premium	How do artist counts compare across subscription types?
Genre by age group	Age bands and annotated heatmap	How does genre mix differ across age bands?
Subscription split	Group count by subscription type	How many records are Free vs Premium?

5. Descriptive Snapshot

The notebook's **describe()** output reports the following numeric summary statistics across 5,000 records:

Metric	Mean	Median	Minimum	Maximum
Age (years)	36.66	37	13	60
Minutes streamed per day	309.24	316	10	600
Number of songs liked	253.52	254	1	500
Discover Weekly engagement (%)	50.30	50.42	10.02	89.99
Repeat song rate (%)	42.39	41.96	5.00	79.99

Subscription counts in the notebook: Free = 2,474 and Premium = 2,526 (approximately 49.5% and 50.5% of records, respectively). These are record counts, not proof of market share.

6. Potential Business Applications

The analysis can support exploratory discussions about audience segmentation and product engagement. Possible follow-on questions include:

• Whether listening duration differs across subscription types or age groups.

• Which genres or artists appear frequently within different listener segments.

• Whether Discover Weekly engagement is associated with repeat-song behaviour.

• Which platform and time-of-day segments may warrant deeper investigation.

• How engagement indicators vary by country, while accounting for sample composition.

These are proposed next steps, not findings already established by the current notebook.

7. Skills Demonstrated

Technical	Python; Pandas; NumPy; Matplotlib; Seaborn; Jupyter Notebook; exploratory data analysis; descriptive statistics; categorical analysis; data-quality checks; data visualization.
Analytical	Question framing; segmentation; interpreting distributions; comparing groups; communicating patterns and limitations.

8. Limitations & Next Steps

• This is descriptive EDA; it does not establish causation or predict future behaviour.

• The notebook references a CSV input file. Include the dataset or document the source and setup steps in the repository so reviewers can reproduce the analysis.

• The current notebook includes visualizations, but the available saved outputs do not expose all exact category counts; therefore, this report avoids naming a single “most popular” platform or genre.

• Next steps could include a reproducible README, clearer chart interpretations, statistical comparisons, and a dashboard for stakeholder-facing exploration.

Recruiter takeaway

This project demonstrates a practical EDA workflow: inspect a dataset, check data quality, summarize numerical variables, visualize categorical distributions, and compare listener segments to generate business questions.

Project artifact: Jupyter Notebook — Capstone_Project_(Global_Music_Streaming_Listener_Preferences).ipynb